

Post-Disaster Rescue Guidance via VLM Fine-Tuning

CSE499A – Section 15, Group 5

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Proposed Solution Report: Cost-Effective, Fine-Tuned VLMs for Disaster Damage Assessment

Abstract—Recent advancements in Large Vision-Language Models (LVLMs) show potential for post-disaster analysis, yet current systems face a severe dichotomy. Open-source models lack disaster-specific domain knowledge, while multi-stage pipelines and commercial orchestrators suffer from extreme computational bloat, hallucinations, and high costs. We propose a Two-Tiered Hybrid Framework: a locally fine-tuned Qwen2.5-VL-7B (optimized via 4-bit QLoRA on a single T4 GPU) for precise edge-capable damage assessment, coupled with DisasTeller, a multi-agent orchestration framework. By abandoning restrictive image-cropping and zero-shot VLM juries, this framework achieves semantically aware, end-to-end automation of public alerting, resource deployment, and recovery planning without relying on heavy compute clusters.

Index Terms—LVLM Fine-Tuning, QLoRA, Multi-Agent Orchestration, Post-Disaster Analysis, Semantic Reasoning, Edge Computing

I. INTRODUCTION AND MOTIVATION

In the aftermath of catastrophic natural events, rapid coordinated decision-making is vital. While automated AI pipelines have been proposed, an in-depth review reveals profound architectural gaps. Dietrich et al. [1] attempted semantic segmentation on satellite data, but building localization completely failed during sparse incidents. Standard segmentation lacks the high-level semantic reasoning required to assess structural collapse.

To compensate, Shakya et al. [2] proposed a computationally heavy three-stage framework (VRT + YOLO + zero-shot VLM). However, rigid image cropping strips the VLM of vital surrounding context (e.g., adjacent debris or floodwaters). Furthermore, relying on unadapted foundation models necessitates a slow, multi-model “VLM-as-a-Jury” validation loop to suppress hallucinations, creating an extreme footprint impossible to deploy on edge hardware. Additionally, standard multi-modal repositories suffer from severe class imbalances [3], [4], causing generic models to overfit to residential structures, while initial perception errors trigger fatal cascading hallucinations in downstream orchestrators [5].

We bridge this gap via a unified, resource-efficient architecture that embeds disaster-specific domain knowledge directly into the weights of a lightweight model, bypassing external VLM juries and destructive crop filters.

II. PROPOSED METHODOLOGY

Our framework resolves operational bottlenecks through two synergistic layers: an Edge Perception Tier and an Orchestration Tier.

A. Tier 1: Resource-Efficient VLM Fine-Tuning

To suppress hallucination risks, we fine-tune Qwen2.5-VL-7B using 4-bit NF4 QLoRA.

1) Eliminating Cropping & Juries

Unlike [2], we utilize native Dynamic Resolution to pass whole-scene high-resolution frames directly into the system. This preserves macro-spatial context, allowing the model to reason about structural damage in relation to its environment. Specialized domain knowledge baked into the neural layers ensures reliable output without secondary validation juries.

2) Stratified Sharding & Hardware Democratization

Addressing imbalances [4], we implement stratified sharding across DisasterM3 instruction subsets to evenly distribute destruction profiles across all functional building classes. We migrate training to a single commodity T4 GPU. By applying gradient accumulation and restricting multi-modal input to a $512 \times 28 \times 28$ matrix, we prevent Out-Of-Memory (OOM) faults. We freeze the vision tower to preserve geospatial feature extraction, applying trainable LoRA matrices solely to the LLM’s linear layers.

3) Super-Resolution Preprocessing

Following [2], we retain a Video Restoration Transformer (VRT) as an optional preprocessing bridge for blurred imagery. A script converts a static frame into a REDS-style pseudo-video sequence. Inference is executed with spatial padding tiles (256×256), allowing high-resolution output reconstruction without enterprise GPUs.

B. Tier 2: DisasTeller Multi-Agent Orchestration

We translate structured visual indicators into actionable logistics via DisasTeller, a CrewAI-based orchestrator powered by Gemini 3.1 Flash-Lite. By feeding it well-formed Tier 1 JSONs, we mitigate cascading planning failures [5]. It features four sequential agents:

- **Expert Team:** Ingests localized assessments and queries humanitarian metrics (EMS-98, FEMA ESF) to finalize destruction grades.
- **Alerts Team:** Correlates metadata to construct hazard boundaries and broadcast targeted public safety alerts.
- **Emergency Service:** Maps optimal zones for temporary shelters, triage stations, and supply routes.
- **Assignment Team:** Outlines human resource deployments and preliminary reconstruction budgets.

TABLE I
COMPARATIVE ANALYSIS OF ARCHITECTURAL APPROACHES

Metric	Traditional Segmentation [1]	Multi-Stage Zero-Shot Jury [2]	Proposed Two-Tiered Framework
Perception Engine	Geospatial Foundation / U-Nets	VRT + YOLOv11 + Zero-Shot VLMs	Fine-tuned Qwen2.5-VL-7B (QLoRA)
Computational Load	Low to moderate server use	Extreme bloat from $4\times$ large VLM juries	Ultra-low; single T4 edge-capable GPU
Context Preservation	Limited pixel-level matching	Degraded by destructive image cropping	Whole-scene native dynamic resolution
Structural Reasoning	Absent; no semantic depth	Fragmented by zero-shot restrictions	Deep domain knowledge baked in weights
Logistical Integration	None; stops at segmentation masks	Basic textual report generation only	End-to-end autonomous command orchestration

The orchestrator enforces a global limit of two Requests Per Minute (RPM = 2), maintaining strategic coordination without expensive enterprise access.

[5] L. Chen et al., “Integration of large vision language models for efficient post-disaster damage assessment,” *JPDAS*, 2026.

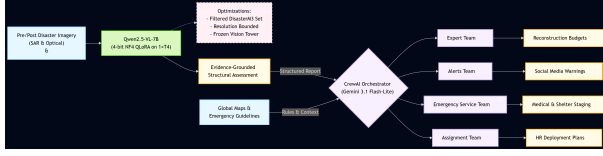


Fig. 1. Proposed Two-Tiered Hybrid Framework Pipeline.

III. EVALUATION & EXPECTED RESULTS

We replace expensive subjective assessments with strict deterministic verification. Perception capabilities are evaluated via hard-coded Multiple-Choice Question (MCQ) accuracy and JSON key-value patterns. We expect a 10-15% accuracy increase over zero-shot baselines in tasks like Scene Recognition and Severity Assessment.

For the command layer, success is measured by the autonomous generation of six MIRA-compliant briefs: (1) Damage Assessment, (2) Social Media Alerts, (3) Medical Staging Protocol, (4) HR Allocation, (5) Public Advisories, and (6) Reconstruction Action Plan.

IV. DEPLOYMENT & CONCLUSION

Our architecture democratizes emergency AI. The 4-bit Qwen2.5-VL-7B footprint runs locally on edge servers in offline crisis zones. When networks partially restore, the generated JSON outputs sync with the Tier 2 orchestrator using minimal bandwidth, matching satellite phone data limits via the strict RPM = 2 constraint.

Ultimately, our framework overcomes the semantic limits of segmentation and the computational bloat of VLM juries, providing a scalable, cost-efficient architecture for practical disaster response. Future work will expand perception to UAV video feeds and simulate real-time communication drops.

REFERENCES

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